CHAPTER 20

Probability

Generally this chapter is considered as an extension of permutation and combination, since it is as logical in nature as permutation and combination. I hope that you must have been well-versed with the permutation and combination (previous chapter) which will enable you to solve the problems of this chapter quickly and easily.

This chapter is important for XAT, SNAP and other MBA entrance tests. Over the past years CAT has shown no great interest in this chapter, still you should learn it heartly due to uncertain syllabus and irregular pattern of CAT.

Remember that so far CAT has asked very simple and logical problems from this chapter which could be solved by applying just commonsense. The literal meaning of probability is the chance of occurrence of an event. For example, if a person is standing at the crossing of two roads which direct towards North, South, East and West.

Thus he has total four alternatives (*i.e,* four different directions) to proceed. Now if he wish to go towards a particular direction then the probability of completing his wish is 1 / 4 since he can choose only one direction at a time out of four directions.

Consider another example, A person has two different cars *viz.,* Scorpio and Safari, which he uses randomly,then it can be said that the probability of using Scorpio is 1 / 2 since out of two cars he can use any one car at a time. Similarly the probability of using Safari is also $\frac{1}{2}$. Thus we can say that the probability of using any one car at a time is 1 / 2 *i.e.,* 50%. So there are chances that in 50% cases he can use Scorpio and in other 50% cases he can use Safari.

Hence from the above illustrations we can conclude that the probability of an event

$$= \frac{(\text{No. of ways in which favourable (or desired event occurs)})}{(\text{Total number of possible outcomes})}$$

Chapter Checklist

- *Important Definitions*
- *Probability*
- *Important Values*
- *Important Addition Theorem*
- *Conditional Probability*
- *Multiplication Theorem*
- *Independent Events*
- *Law of Total Probability*
- *Baye's Rule*
- ***CAT Test***

20.1 Important Definitions

Experiment

An operation which can produce some well-defined outcomes, is known as an experiment.

The various experiments, when repeated under identical conditions, results. (*i.e.,* outcomes) in each case are same *e.g.,* standard scientific experiments.

But there are some other experiment, when repeated under identical conditions, results in each case are different *e.g.,* rolling of a fair die, tossing of a coin etc.

Random Experiment

If in each trial of an experiment conducted under identical conditions, the outcome is not unique, but may be any of the possible outcomes then such an experiment is known as a random experiment. *e.g.,* rolling of an unbiased die, tossing of a fair coin, drawing of a card from a well shuffled pack of cards.

Sample Space

The set of all possible outcomes in a random experiment is called a sample space and it is generally denoted by S.

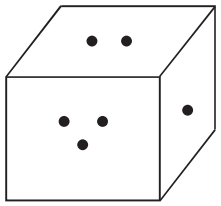
If $E_1, E_2, E_3 \ldots E_n$ are the possible outcomes of a random experiment, then $S = \{E_1, E_2, \ldots . E_n\}$. Also, each of the element of sample space 'S' is called a **sample point.**

***Exp.* 1)** In tossing of a fair coin, there are two possible outcomes, viz., head (H) and tail (T). So, the sample space in this random experiment is $S = \{H, T\}$

***Exp.* 2)** When two fair coins are tossed together, the possible outcomes of the experiment are HH, HT, TH and TT. So the sample space is given by $S = \{HH, HT, TH, TT\}$

***Exp.* 3)** When unbiased die is thrown, it gives 6 possible outcomes viz., 1,2, 3, 4, 5 and 6. So, the sample space $S = \{1, 2, 3, 4, 5, 6\}$

NOTE A die is a cubical solid having 6 similar faces. In each of the six faces there are unique number of holes *viz.,* 1 hole, 2 holes, 3 holes, 4 holes, 5 holes and 6 holes.
Remember that the sum of number of holes in any two opposite faces is always 7 *viz.,* $1+6 = 2+5 = 3+4 = 7$.



- Die means one (single) die
- Dice means more than one die.
- In throwing a die, the outcome is the number of holes on the uppermost face

***Exp.* 4)** When two unbiased dice are rolled (or tossed) simultaneously then there are total $6 \times 6 = 36$ possible outcomes. So, the sample space

$$S = \begin{Bmatrix} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1) \ldots\ldots\ldots\ldots\ldots\ldots\ldots (3,6) \\ (4,1) \ldots\ldots\ldots\ldots\ldots\ldots\ldots (4,6) \\ (5,1) \ldots\ldots\ldots\ldots\ldots\ldots\ldots (5,6) \\ (6,1) \ldots\ldots\ldots\ldots\ldots\ldots\ldots (6,6) \end{Bmatrix}$$

***Exp.* 5)** When a die and a coin are tossed simultaneously, then there are total 12 possible outcomes. So the sample space

$$S = \begin{Bmatrix} (1,H), (2,H), (3,H), (4,H), (5,H), (6,H) \\ (1,T), (2,T), (3,T), (4,T), (5,T), (6,T) \end{Bmatrix}$$

***Exp.* 6)** A coin is tossed twice. If the second throw results in a tail, then a die is thrown. So in this random experiment sample space.

$$S = \begin{Bmatrix} HH, TH, HT1, HT2, HT3, HT4, HT5, HT6, \\ TT1, , TT2, , TT3, TT4, TT5, TT6, \end{Bmatrix}$$

***Exp.* 7)** From a bag containing 2 black and 3 white balls we draw two balls.

Let B_1, B_2 be the black balls and W_1, W_2 and W_3 be the white balls then the sample space

$$S = \begin{Bmatrix} B_1W_1, & B_1W_2, & B_1W_3 \\ B_2W_1, & B_2W_2, & B_2W_3 \\ B_1B_2, & W_1W_2, & W_2W_3 \\ W_1W_3 & & \end{Bmatrix}$$

Event

Any subset of a sample space is called an event.

Exp. 1) In a single throw of a die, the event of getting an even number is given by $E = \{2, 4, 6\}$

Clearly, here the sample space $S = \{1, 2, 3, 4, 5, 6\}$

Hence $E \subset S$ *i.e.*, E is the subset of S.

Exp. 2) Consider a random experiment of tossing two dice at a time. The sample space $S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5) \ldots (6, 1), (6, 2) \ldots (6, 6)\}$. Some different events associated with the above sample space are given below.

$E_1 = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$

$E_2 = \{(1, 8), (2, 7), (3, 6), (4, 5), (5, 4), (6, 3), (7, 2), (8, 1)\}$

$E_3 = \{(1, 2), (1, 4), (1, 6), (3, 2), (3, 4), (3, 6), (5, 2), (5, 4), (5, 6)\}$

$E_4 = \{(2, 2), (2, 3), (2, 5), (3, 2), (3, 3), (3, 5), (5, 2), (5, 3), (5, 5)\}$ etc.

Clearly $E_1 \subset S, E_2 \subset S, E_3 \subset S, E_4 \subset S$

Where E_1 is the event of getting a doublet and E_2 is the event of getting 9 as the sum. E_3 is the event of getting odd number on the first die and an even number on the second die. E_4 is the event of getting only prime number on each of the two dice.

Exp. 3) Consider a random experiment of tossing three coins at a time. The sample space S = {HHH, HHT, HTH, THH, TTH, HTT, THT, TTT}

Following are the various events associated with the above sample space.

$E_1 = \{HHH, HHT, HTH, THH\}$

$E_2 = \{HHT, HTH, THH, TTH, HTT, THT\}$

$E_3 = \{HHH, TTT\}$

$E_4 = \{THH, TTH, THT, TTT\}$

Where E_1 is the event of getting atleast two heads and E_2 is the event of getting atleast one head and atmost two heads.

E_3 is the event of getting all the three coins identical and E_4 is the event of getting tail on the first coin.

Elementary Events An event containing only a single sample point is called an elementary event, or simple event.

Exp. 1) In a simultaneous toss of two coins the sample space $S = \{HH, HT, TH, TT\}$

then $E_1 = \{HH\}$

$E_2 = \{TT\}$ are the elementary events.

Exp. 2) Consider a random experiment in which two dice rolled simultaneously, the sample space

$S = \{(1, 1), (1, 2) \ldots (6, 5), (6, 6)\}$

then $E_1 = \{(1, 1)\}$

$E_2 = \{(6, 6)\}$

$E_3 = \{(5, 5)\}$ are the elementary events.

Here E_1 is the event of getting the sum of two.

E_2 is the event of getting the sum of twelve

E_3 is the event of getting the product of 25.

Compound Events Events which are not elementary are known as compound events or the events which contains more than one element are called compound or composite events.

Exp. 1) Consider a random experiment in which two dice are rolled simultaneously, the sample space

$S = \{(1, 1), (1, 2) \ldots\ldots (6, 5), (6, 6)\}$.

then $E_1 = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$

$E_2 = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$

$E_3 = \{(1, 2), (2, 4), (3, 6)\}$

are the compound events.

Here E_1 is the event of getting the sum of 6

E_2 is the event of getting identical results

E_3 is the event of getting the twice number by the second die than that by first die.

Occurrence of Events In a random experiment, let S be the sample space and E be the event such that $E \subseteq S$, Let w be an outcome of a trial such that $w \in A$, then we say that the event E has occurred. If $w \notin E$, we say that the event E has not occurred.

Exp. 1) Consider the random experiment of throwing an unbiased die. Let E be the event of getting an odd number, then $E = \{1, 3, 5\}$

Now, in a trial, let the outcome be 3, Since $3 \in E$, so in this trial, the event E has occurred. In another trial, let the outcome be 4.

Since $4 \notin E$ so in this trial, the event E has not occurred.

Exp. 2) Consider the random experiment of throwing a die. Let the outcome of the trial is 6. Then we can say that the following events have occurred.

(i) Getting a number greater than or equal to 3, represented by the set {3, 4, 5, 6}

(ii) Getting an even number represented by the set {2, 4, 6}

We can also say that the following events have not occurred

(i) Getting an odd number represented by the set {1, 3, 5}.

(ii) Getting a prime number represented by the set {2, 3, 5}

Impossible Events Let S be a sample space associated with a random experiment, Now, since $\phi \subseteq S$, so ϕ is an event, called an impossible event.

Sure (or certain) **Event** Let S be a sample space associated with a random experiment. Now, since $S \subseteq S$, so S is an event, called a sure or certain event.

Example Consider the random experiment of throwing an unbiased die. The sample space $S = \{1, 2, 3, 4, 5, 6\}$.

Let E_1 be the event of getting a number less than 1.

and E_2 be the event of getting a number greater than 6.

Here E_1 and E_2 both are impossible events.

Let E_3 be the event of getting a number less than or equal to 7. and E_4 be the event of getting a number multiple of 2 but less than 7. Here E_3 and E_4 are the certain events.

Equally Likely Events Events are said to be equally likely, if none of them is expected to occur in preference to the other.

Example If an unbiased die is rolled, then each outcome is equally likely to happen i.e., all elementary events are equally likely to happen. If however, the die is so formed that a particular face occurs most often, then the die is biased. So in this case, the outcomes are not equally likely to happen.

Favourable Events Let S be the sample space associated with a random experiment and let $E \subset S$. Then the elementary events belonging to E are known as the favourable events to A.

***Exp.* 1)** Consider a random experiment of throwing a die. Let E be the event of getting an even number, then $E = \{2, 4, 6\}$. So there are three favourable events of event E *viz.* {2}, {4} and {6}.

***Exp.* 2)** Consider a random experiment of throwing a pair of dice. Let E be the event of getting the sum as 8, then $E = \{(1, 7), (2, 6), (3, 5), (4, 4), (5, 3), (6, 2), (7, 1)\}$

So, there are 7 favourable events to event E.

Complementary Events In a random experiment, let S be the sample space and let E be an event. Then $E \subseteq S$. Clearly, $E^c \subseteq S$. So E^c is also an event, called the complementary of E. Sometimes E^c is denoted by $\overline{E}$ or E'. Where $\overline{E}$ is called as "not-E". Thus it is clear that $\overline{E}$ occurs only when E does not occur. Also in a trial one and only of E and $\overline{E}$ must occur.

Algebra of Events

In a random experiment, let the sample space be S. Let $E \subseteq S$, $F \subseteq S$ and $G \subseteq S$, then we can define the following relations in the set theory mode.

(i) $\overline{E}$ is an event that occurs only when E does not occur.

(ii) $(E \cup F)$ is an event that occurs only when E occurs or F occurs or both occurs.

(iii) $(E \cap F)$ is an event that occurs only when each one of E and F occurs.

(iv) $(E \cap \overline{F})$ is an event that occurs only when E occurs but not F.

(v) $(\overline{E} \cap \overline{F}) = (\overline{E \cup F})$ is an event that occurs when neither E nor F occurs.

(vi) $(E \cup F \cup G)$ is an event that occurs when at least one of E, F or G occurs.

(vii) $(E \cap F \cap G)$ is an event that occurs when all three E, F and G occurs.

(viii) $(E \cap \overline{F}) \cup (\overline{E} \cap F)$ is an event that occurs when exactly one of E and F occurs.

(ix) $(E \cap F \cap \overline{G}) \cup (E \cap \overline{F} \cap G) \cup (\overline{E} \cap F \cap G)$ is an event that occurs when exactly two of E, F and G occurs.

Mutually Exclusive Events

Let S be the sample space associated with a random experiment and let E_1 and E_2 be the two events. Then E_1 and E_2 are mutually exclusive events if $E_1 \cap E_2 = \phi$

***Exp.* 1)** Consider a random experiment of throwing a die. Let E_1, E_2 and E_3 be three events such that

$E_1 = \{1, 3, 5\}$, the event of getting an odd number

$E_2 = \{2, 4, 6\}$, the event of getting an even number

$E_3 = \{2, 3, 5\}$, the event of getting a prime number.

Clearly $E_1 \cap E_2 = \phi$

and $E_1 \cap E_3 \neq \phi, E_2 \cap E_3 \neq \phi$

Hence E_1 and E_2 are mutually exclusive events but E_1 and E_3, E_2 and E_3 are not mutually exclusive.

Mutually Exclusive & Exhaustive System of Events

Let S be the sample space associated with a random experiment.

Let $E_1, E_2, \ldots, E_n$ be the subsets of S such that

(i) $E_i \cap E_j = \phi$ for $i \neq j$ and

(ii) $E_1 \cup E_2 \cup E_3 \cup \ldots \cup E_n = S$

then the set of events $E_1, E_2, E_3, \ldots, E_n$ is said to form a mutually exclusive and exhaustive system of events.

***Exp.* 1)** In a sample space all the elementary events form a mutually exclusive and exhaustive system of events.

***Exp.* 2)** Let $S = \{1, 2, 3, 4, 5, 6\}$ be the sample space when an unbiased die is rolled.

Let $E_1 = \{1, 3, 5\}$ and $E_2 = \{2, 4, 6\}$, then E_1 and E_2 form the mutually exclusive and exhaustive system of events.

Also, let $E_3 = \{1, 2\}$, $E_4 = \{4, 5\}$, then E_3 and E_4 do not form the mutually exclusive and exhaustive system of events since E_3 and E_4 do not include all the elementary events of S.

Again if $E_5 = \{3, 6\}$ and $E_6 = \{2, 4, 6\}$, then E_5 and E_6 do not form mutually exclusive and exhaustive system of events since $E_5 \cap E_6 \neq \phi$ and E_5, E_6 do not include all the elementary events of the sample space S. A pack of cards consists of 52 cards. There are four suits each containing 13 cards called as spades, clubs, hearts and diamonds.

All the spades and clubs are black faced cards while hearts and diamonds are red faced cards. The aces, kings, queens and jacks are known as face cards.

In each suit there is one ace, one king, one queen and one jack and rest 9 cards are numbered cards.

20.2 Probability

In a random experiment, let S be the sample space and let $E \subseteq S$. where E is an event.

The probability of occurrence of the event E is defined as

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{number of possible outcomes}}$$

$$= \frac{\text{number of elements in } E}{\text{number of elements in } S} = \frac{n(E)}{n(S)}$$

$$= \frac{\text{number of elementary events in } E}{\text{number of elementary events in } S}$$

From the above definitions it is clear that

(i) $0 \le P(E) \le 1$

(ii) $P(\phi) = 0$

(iii) $P(S) = 1$

Also, $$P(\overline{E}) = \frac{\text{number of elementary events in } \overline{E}}{\text{number of elementary events in } S}$$

$$= \frac{n(S) - n(E)}{n(S)}$$

$$= 1 - \frac{n(E)}{n(S)} = 1 - P(E)$$

$\Rightarrow$ $$P(\overline{E}) = 1 - P(E)$$

$\therefore$ $$P(E) + P(\overline{E}) = 1$$

Odds in Favour of An Event and Odds Against An Event

In m be the number of ways in which an event occurs and n be the number of ways in which it does not occur, then

(i) odds in favour of the event $= \frac{m}{n}$ (or $m : n$)

(ii) odds against the event $= \frac{n}{m}$ (or $n : m$)

Introductory Exercise 20.1

1. Find the probability of getting a head in a throw of a coin.

(a) $\frac{1}{2}$ (b) 1

(c) 2 (d) none of these

Directions (for Q. Nos. 2 to 5) *Two fair coins are tossed simultaneously. Find the probability of*

2. Getting only one head.

(a) 1/2 (b) 1/3 (c) 2/3 (d) 3/4

3. Getting atleast one head.

(a) $\frac{1}{4}$ (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{3}{8}$

4. Getting two heads.

(a) $\frac{2}{7}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) $\frac{4}{5}$

5. Getting atleast two heads :

(a) $\frac{3}{4}$ (b) $\frac{1}{2}$

(c) $\frac{1}{4}$ (d) 1

Directions (for Q. Nos. 6 to 12) *Three fair coins are tossed simultaneously. Find the probability of*

6. Getting one head.

(a) 0 (b) 3/4 (c) $\frac{5}{8}$ (d) $\frac{3}{8}$

7. Getting one tail.

(a) 1 (b) $\frac{1}{4}$ (c) $\frac{5}{8}$ (d) $\frac{3}{8}$

8. Getting atlest one head.

(a) $\frac{7}{8}$ (b) $\frac{1}{8}$ (c) $\frac{3}{4}$ (d) $\frac{1}{4}$

9. Getting two heads.

(a) $\frac{3}{5}$ (b) $\frac{3}{8}$ (c) $\frac{5}{8}$ (d) $\frac{2}{5}$

10. Getting atleast two heads.

(a) $\frac{3}{8}$ (b) $\frac{7}{8}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

11. Getting atleast one head and one tail.

(a) $\frac{2}{8}$ (b) $\frac{1}{2}$ (c) $\frac{3}{10}$ (d) $\frac{3}{4}$

12. Getting more heads than the number of tails.
(a) 2 (b) 7/8 (c) $\frac{5}{8}$ (d) $\frac{1}{2}$

Directions (for Q. Nos. 13 to16) *An unbiased die is rolled. Find the probability of*

13. Getting a number less than 7 but greater than zero.
(a) 0 (b) 3/4 (c) 1 (d) $\frac{7}{8}$

14. Getting a multiple of 3.
(a) $\frac{1}{6}$ (b) $\frac{1}{3}$
(c) $\frac{5}{6}$ (d) none of these

15. Getting a prime number.
(a) $\frac{1}{2}$ (b) $\frac{3}{5}$ (c) $\frac{5}{7}$ (d) $\frac{5}{8}$

16. Getting an even number.
(a) $\frac{1}{2}$ (b) $\frac{4}{5}$ (c) $\frac{2}{8}$ (d) $\frac{3}{4}$

Directions (for Q. Nos. 17 and 18): *A coin is tossed successively three times. Find the probability of*

17. Getting exactly one head or two heads.
(a) $\frac{1}{4}$ (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{3}{8}$

18. Getting no heads.
(a) 0 (b) 1 (c) $\frac{1}{8}$ (d) $\frac{7}{8}$

Directions (for Q. Nos. 19 to 27) *Two dice are rolled simultaneously.Find the probability of*

19. Getting a total of 9.
(a) $\frac{1}{3}$ (b) $\frac{1}{9}$ (c) $\frac{8}{9}$ (d) $\frac{9}{10}$

20. Getting a sum greater than 9.
(a) $\frac{10}{11}$ (b) $\frac{5}{6}$ (c) $\frac{1}{6}$ (d) $\frac{8}{9}$

21. Getting a total of 9 or 11.
(a) $\frac{2}{99}$ (b) $\frac{20}{99}$ (c) $\frac{1}{6}$ (d) $\frac{1}{10}$

22. Getting a doublet.
(a) 1/12 (b) 0 (c) 5/8 (d) 1/6

23. Getting a doublet of even numbers.
(a) 5/8 (b) 1/12 (c) 3/4 (d) 1/4

24. Getting a multiple of 2 on one die and a multiple of 3 on the other.
(a) $\frac{15}{36}$ (b) $\frac{25}{36}$ (c) $\frac{11}{36}$ (d) $\frac{5}{6}$

25. Getting the sum of numbers on the two faces divisible by 3 or 4.
(a) $\frac{4}{9}$ (b) $\frac{1}{7}$ (c) $\frac{5}{9}$ (d) $\frac{7}{12}$

26. Getting the sum as a prime number.
(a) $\frac{3}{5}$ (b) $\frac{5}{12}$ (c) $\frac{1}{2}$ (d) $\frac{3}{4}$

27. Getting atleast one '5'.
(a) $\frac{3}{5}$ (b) $\frac{1}{5}$ (c) $\frac{5}{36}$ (d) $\frac{11}{36}$

Directions (for Q. Nos. 28 to 35) *One card is drawn from a pack of 52 cards, each of the 52 cards being equally likely to be drawn. Find the probability that*

28. The card drawn is black.
(a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) $\frac{8}{13}$ (d) can't be determine

29. The card drawn is a queen.
(a) $\frac{1}{12}$ (b) $\frac{1}{13}$
(c) $\frac{1}{4}$ (d) $\frac{3}{4}$

30. The card drawn is black and a queen.
(a) $\frac{1}{13}$ (b) $\frac{1}{52}$ (c) $\frac{1}{26}$ (d) $\frac{5}{6}$

31. The card drawn is either black or a queen.
(a) $\frac{15}{26}$ (b) $\frac{13}{17}$ (c) $\frac{7}{13}$ (d) $\frac{15}{26}$

32. The card drawn is either king or a queen.
(a) $\frac{5}{26}$ (b) $\frac{1}{13}$ (c) $\frac{2}{13}$ (d) $\frac{12}{13}$

33. The card drawn is either a heart, a queen or a king.
(a) $\frac{17}{52}$ (b) $\frac{21}{52}$ (c) $\frac{19}{52}$ (d) $\frac{9}{26}$

34. The card drawn is neither a spade nor a king.
(a) 0 (b) $\frac{9}{13}$ (c) $\frac{1}{2}$ (d) $\frac{4}{13}$

35. The card drawn is neither an ace nor a king.
(a) $\frac{11}{13}$ (b) $\frac{1}{2}$ (c) $\frac{2}{13}$ (d) $\frac{11}{26}$

36. From a well shuffled pack of 52 cards, three cards are drawn at random. Find the probability of drawing an ace, a king and a jack.
(a) $\frac{16}{5525}$ (b) $\frac{16}{625}$
(c) $\frac{16}{3125}$ (d) none of these

37. Four cards are drawn at random from a pack of 52 cards. Find the probability of getting all the four cards of same number.
(a) $\frac{17}{1665}$ (b) $\frac{1}{20825}$
(c) $\frac{7}{25850}$ (d) none of these

38. From a well shuffled pack of 52 playing cards, four cards are accidently dropped. Find the probability that one card is missing from each suit.
(a) $\frac{17}{20825}$ (b) $\frac{2197}{20825}$
(c) $\frac{197}{1665}$ (d) none of these

39. Four cards are drawn at random from a pack of 52 cards. Find the probability of getting all the four cards of different numbers.
(a) $\frac{141}{4165}$ (b) $\frac{117}{833}$
(c) $\frac{264}{4165}$ (d) none of these

Directions (for Q. Nos. 40 to 43) *Four dice are thrown simultaneously. Find the probability that*

40. All of them show the same face.
(a) $\frac{1}{216}$ (b) $\frac{15}{16}$ (c) $\frac{15}{36}$ (d) $\frac{1}{2}$

41. All of them show the different face.
(a) $\frac{3}{28}$ (b) $\frac{5}{18}$ (c) $\frac{15}{36}$ (d) $\frac{11}{36}$

42. Two of them show the same face and remaining two show the different faces.
(a) $\frac{4}{9}$ (b) $\frac{5}{9}$ (c) $\frac{11}{18}$ (d) $\frac{7}{9}$

43. Atleast two of them show the same face.
(a) $\frac{37}{72}$ (b) $\frac{11}{36}$ (c) $\frac{47}{72}$ (d) $\frac{25}{36}$

44. What is the probability that a number selected from the numbers 1, 2, 3, ..., 20, is a prime number when each of the given numbers is equally likely to be selected?
(a) 7/10 (b) 2/15 (c) 2/5 (d) 3/5

45. Tickets are numbered from 1 to 18 are mixed up together and then 9 ticket is drawn at random. Find the probability that the ticket has a number, which is a multiple of 2 or 3.
(a) $\frac{1}{3}$ (b) $\frac{3}{5}$ (c) $\frac{2}{3}$ (d) $\frac{5}{6}$

46. In a lottery of 100 tickets numbered 1 to 100, two tickets are drawn simultaneously. Find the probability that both the tickets drawn have prime numbers.
(a) $\frac{2}{33}$ (b) $\frac{7}{50}$ (c) $\frac{7}{20}$ (d) $\frac{5}{66}$

47. In the previous question (number 46), find the probability that none of the tickets drawn has a prime number.
(a) $\frac{29}{66}$ (b) $\frac{17}{33}$ (c) $\frac{37}{66}$ (d) $\frac{17}{50}$

48. Find the probability that a leap year selected at random will contain 53 Sundays.
(a) $\frac{5}{7}$ (b) $\frac{3}{4}$ (c) $\frac{4}{7}$ (d) $\frac{2}{7}$

Directions (for Q. Nos. 49 to 53) *A bag contains 8 red and 4 green balls. Find the probability that*

49. The ball drawn is red when one ball is selected at random.
(a) $\frac{2}{3}$ (b) $\frac{1}{3}$ (c) $\frac{1}{6}$ (d) $\frac{5}{6}$

50. All the 4 balls drawn are red when four balls are drawn at random.
(a) $\frac{17}{32}$ (b) $\frac{14}{99}$
(c) $\frac{7}{12}$ (d) none of these

51. All the 4 balls drawn are green when four balls are drawn at random.
(a) $\frac{1}{495}$ (b) $\frac{7}{99}$ (c) $\frac{5}{12}$ (d) $\frac{2}{3}$

52. Two balls are red and one ball is green when three balls are drawn at random.
(a) $\frac{56}{99}$ (b) $\frac{112}{495}$
(c) $\frac{78}{495}$ (d) none of these

53. Three balls are drawn and none of them is red.
(a) $\frac{68}{99}$ (b) $\frac{7}{99}$
(c) $\frac{4}{495}$ (d) none of these

54. The odds in favour of an event are 2 : 7. Find the probability of occurrence of this event.
(a) $\frac{2}{9}$ (b) $\frac{5}{12}$ (c) $\frac{7}{12}$ (d) $\frac{2}{5}$

55. The odds against of an event are 5 : 7, find the probability of occurrence of this event.
(a) $\frac{3}{8}$ (b) $\frac{7}{12}$ (c) $\frac{2}{7}$ (d) $\frac{5}{12}$

56. If there are two children in a family, find the probability that there is atleast one girl in the family.
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{3}{4}$ (d) none of these

57. From a group of 3 men and 2 women, two persons are selected at random. Find the probability that atleast one woman is selected.
(a) $\frac{1}{5}$ (b) $\frac{7}{10}$ (c) $\frac{2}{5}$ (d) $\frac{5}{6}$